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رياضيات 5 + 6

ذكاء ثالثة

الرياضيات المتقطعة العددية

المحاضرة الخامسة والسادسة

المرحلة الثالثة

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🧩 1. Introduction to Graph Theory

Graph Theory is one of the main branches of Discrete Mathematics.
It studies the relationships between objects using **vertices (nodes)** and **edges (connections)**.

◆ Definition:

A graph G is an ordered pair:

$$G = (V, E)$$

where:

- V = set of **vertices (or nodes)**
- E = set of **edges (or arcs)** connecting pairs of vertices

Graphs are widely used in:

- Computer Networks
- Social Networks
- Transportation Systems
- Artificial Intelligence
- Internet and Web Modeling

🕒 2. Types of Graphs

2.1 Undirected Graph

◆ Definition:

An **undirected graph** is a graph where edges have **no direction**.
That is, if there is an edge between vertices u and v, then:

$$(u, v) = (v, u)$$

◆ Example 1:

Let

$$V = \{A, B, C, D\}$$

$$E = \{(A, B), (A, C), (B, D), (C, D)\}$$

Graph illustration:

```
A —— B
|    |
C    D
```

◆ Properties:

- You can move between connected vertices in both directions.
- No arrows are used; edges are simply lines.

2.2 Directed Graph (Digraph)

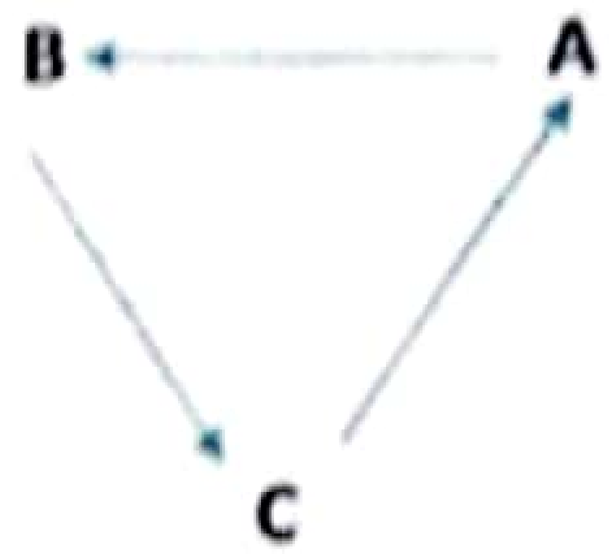
◆ Definition:

A **directed graph** (or **digraph**) is a graph where each edge has a **specific direction.**
So (u, v) means there is an arrow **from u to v,** and it is **not** the same as (v, u).

◆ Example 2:

$$V = \{A, B, C\}, \quad E = \{(A, B), (B, C), (C, A)\}$$

Graph illustration:



يمكن ان تكون على اكثر من شكل

◆ Notes:

- Arrows indicate direction of connection.
- A vertex can have an edge that starts and ends on itself — called a **loop**.

صح المقارنة للاكلاي

3. Comparison: Directed vs Undirected Graph

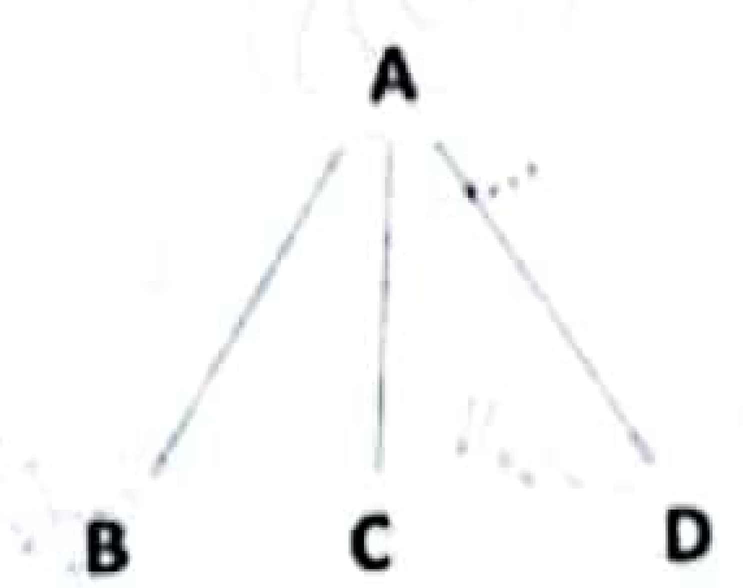
Property	Undirected Graph	Directed Graph
Edge direction	No	Yes
(u,v) = (v,u)?	Yes	No
Representation	Lines	Arrows
Number of possible edges (without loops)	$\frac{n(n-1)}{2}$	$n(n-1)$

4. Degree of a Vertex

In an Undirected Graph:

- The **degree** of a vertex = number of edges incident to it.
- A loop counts as 2 degrees.

Example 3:



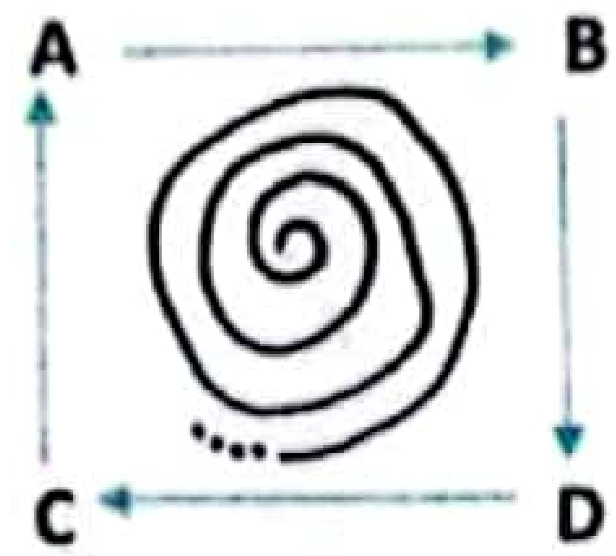
Vertex	Degree
A	3
B	1
C	1
D	1

In a Directed Graph:

Each vertex has **two degrees**:

- **In-degree:** number of edges *coming into* the vertex.
- **Out-degree:** number of edges *going out* of the vertex.

Example 4:



Vertex	الدرجة الداخلية In-degree	الدرجة الخارجية Out-degree
A	1	1
B	1	1
C	1	1
D	1	1

📊 5. Graph Representations

5.1 Adjacency Matrix

If there are n vertices, the **adjacency matrix** is an $n \times n$ matrix A, where each entry $a_{ij} = 1$ if there is an edge from v_i to v_j, and 0 otherwise.

Example 5 (Undirected Graph):

$$V = \{A, B, C\}, \quad E = \{(A, B), (A, C)\}$$

	A	B	C
A	0	1	1
B	1	0	0
C	1	0	0

Example 6 (Directed Graph):

$$E = \{(A, B), (B, C)\}$$

	A	B	C
A	0	1	0
B	0	0	1
C	0	0	0

🍀 6. Special Types of Graphs

1 Simple Graph:

Contains no loops or multiple edges.

2 Multigraph:

Contains multiple edges between the same vertices.

3 Complete Graph (Kn):

Every vertex is connected to all others.

Number of edges = $\frac{n(n-1)}{2}$.

4 Bipartite Graph:

Vertices can be divided into two sets such that no edges connect vertices within the same set.

5 Weighted Graph:

Each edge has a **weight** (cost, distance, or time).

✏️ 7. Solved Examples

Example 9:

Find the number of edges in a complete graph K_5.

$$n = 5$$

$$\text{Number of edges} = \frac{5(5-1)}{2} = 10$$

✅ **Answer:** 10 edges.

Example 10:

For the directed graph:

$$E = \{(A, B), (B, C), (C, A), (A, C)\}$$

Find in-degree and out-degree of each vertex.

Vertex	In-degree	Out-degree
A	1	2
B	1	1
C	2	1

Example 11:

Can we have a simple graph with 4 vertices where each vertex has degree 3?

✅ Yes.

That is a **complete graph** K_4.

$$\text{Number of edges} = \frac{4(4-1)}{2} = 6$$

9. Exercises for Practice

1. Draw an undirected graph with vertices {A, B, C, D, E} where each vertex has degree 2.

2. For the directed graph $E = \{(A,B),(B,C),(C,D),(D,A)\}$, find in-degree and out-degree.

3. Construct the adjacency matrix for:

$$V = \{1,2,3,4\}, \quad E = \{(1,2),(1,3),(3,4)\}$$

المصفوفة المجاورة :-

المقدمة

تُعد مصفوفة المجاورة من أهم طرق تمثيل الرسوم البيانية (Graphs) في الرياضيات المتقطعة.
وهي طريقة رياضية منظمة لتمثيل العلاقات بين الرؤوس في الرسم البياني باستخدام مصفوفة ثنائية الأبعاد (جدول من الأرقام).

تعريف مصفوفة المجاورة

◆ التعريف العام:

إذا كان لدينا رسم بياني $G = (V, E)$ يحتوي على n رأسًا،
فإن مصفوفة المجاورة هي مصفوفة مربعة من الرتبة $n \times n$،
حيث يمثل العنصر a_{ij} العلاقة بين الرأس v_i والرأس v_j.

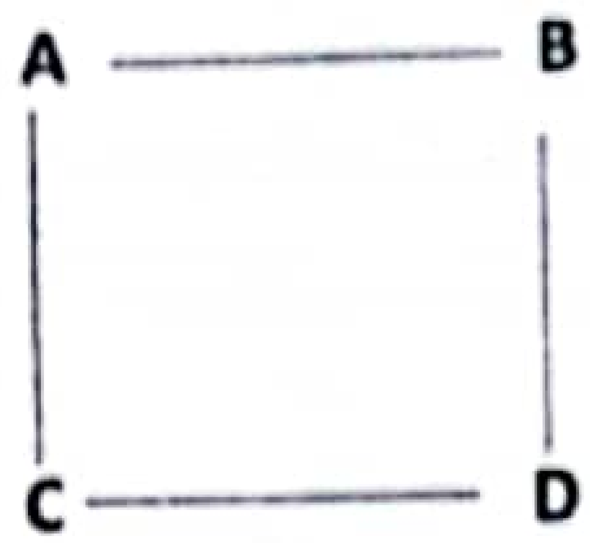
أنواع مصفوفات المجاورة

1 مصفوفة المجاورة للرسم غير الموجّه (Undirected Graph)

- المصفوفة تكون متماثلة (Symmetric) لأن العلاقة بين v_i و v_j هي نفسها بين v_j و v_i.
- لا يوجد اتجاه محدد بين الرؤوس.

◆ مثال 1 (غير موجه):

لدينا الرسم التالي:



$$\{A, B, C, D\} = V$$

$$\{(C, D), (B, D), (A, C), (A, B)\} = E$$

مصفوفة التجاور

الحل:

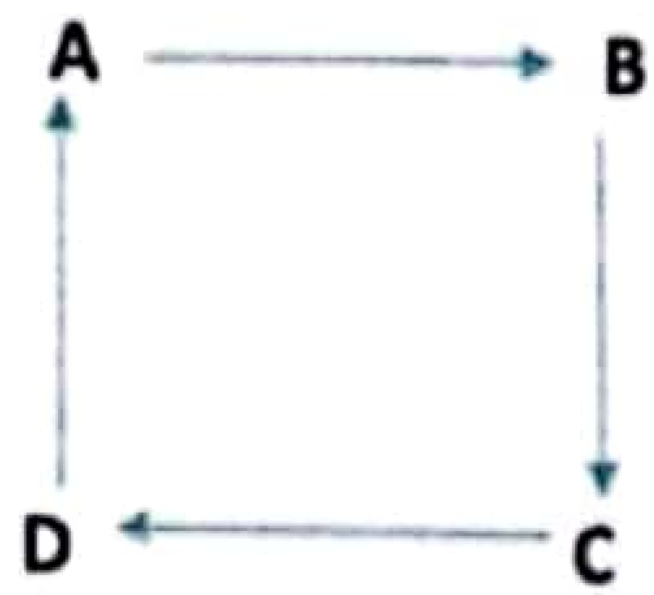
	A	B	C	D
A	0	1	1	0
B	1	0	0	1
C	1	0	0	1
D	0	1	1	0

◆ نلاحظ أن المصفوفة متماثلة حول القطر الرئيسي (A–A, B–B, C–C, D–D).

2 مصفوفة المجاورة للرسم الموجّه (Directed Graph)

- هنا العلاقة بين v_i و v_j تأخذ اتجاهًا محددًا.
- لذلك المصفوفة غير متماثلة بالضرورة.

◆ مثال 2 (موجّه):



$$\{A,B,C\} = V$$

$$\{(C,A),(B,C),(A,B)\} = E$$

الحل:

	A	B	C
A	0	1	0
B	0	0	1
C	1	0	0

◆ لا توجد تماثلية هنا لأن الأسهم لها اتجاه واحد.

◆ مثال 3 (موجه مع عدة أسهم):

$$\{(C,C),(B,A),(A,C),(A,B)\} = E \quad ,\{A,B,C\} = V$$

الحل:

	A	B	C
A	0	1	1
B	1	0	0
C	0	0	1

◆ لاحظ وجود 1 في (C, C) لأنها حلقة ذاتية (Loop) على الرأس C.

ملاحظات مهمة

1. المصفوفة دائمًا مربعة لأن عدد الصفوف = عدد الأعمدة = عدد الرؤوس.
2. القطر الرئيسي (Main Diagonal) يمثل الحلقات الذاتية (Loops).
3. إذا كانت المصفوفة متماثلة ⇐ الرسم غير موجه.
4. إذا لم تكن متماثلة ⇐ الرسم موجه.